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Force limiting: when the shaker is too strong



Learning objective
Explain an acceleration-controlled test overtest using mechanical impedance, calculate a force-induced notch, and judge whether that notch has a defensible qualification basis.

The shaker follows the specified PSD. The control accelerometers look excellent. Yet the test article sees a base reaction that its flight support would not deliver. The missing variable is the mechanical interaction at the mounting interface.

For resonant hardware, holding acceleration constant can require a large force. A flight support deforms and participates in coupled modes; an acceleration-controlled shaker can sustain the commanded motion against a much larger reaction.

Force limiting adds an independent load constraint. The controller reduces input at frequencies where the measured interface force would otherwise exceed its approved limit.

Opening photo: the horizontal shaker built for Webb at NASA Goddard, during construction; the vertical system was not yet installed. Credit: NASA/Chris Gunn. NASA, February 13, 2017 [4]. The photo illustrates test scale; the example below is not Webb test data.

The engineering objective is credible exposure with an intentional margin. Excessive test force can damage good hardware, consume fatigue life, or drive unnecessary mass into a design. An unjustified notch can hide a real weakness. Both errors reduce the value of qualification evidence.

Today’s scope

We use a single translational interface and one linear mode to make the bookkeeping transparent. The numerical force cap is deliberately assumed, not recommended. Multi-point mounting, rotational response and acoustic excitation require additional checks before this example can inform a real test.

Acceleration alone does not define the delivered load

Mechanical impedance is complex force divided by velocity. Apparent mass is force divided by acceleration. For harmonic motion, $M_a(\omega) = Z(\omega)/(j\omega)$. It has units of mass but is a frequency-response function, not a scale reading. Near a resonance, its magnitude can greatly exceed the physical mass; its phase matters when subsystems are coupled.

Consider a mass m connected to a moving base y by stiffness k and viscous damping c. Let x be absolute mass displacement and use an $\exp(j\omega t)$ convention. The force transmitted from the base to the mass is $k(y - x) + c(\dot{y} - \dot{x})$. From equilibrium, this equals $m\ddot{x}$. Therefore:

$$M_a = F/a_b = m(k + j\omega c)/(k - m\omega^2 + j\omega c)$$
$$M_a/m = (1 + j2\zeta r)/(1 - r^2 + j2\zeta r), \quad r = f/f_n$$

At low frequency, the mass follows the base and M_a tends to m. At resonance, $|M_a| = m\sqrt{(1 + 4\zeta^2)}/(2\zeta)$, approximately mQ for small damping. Holding the same base acceleration at that frequency produces a much larger reaction than the rigid-body estimate ma_b. This is an exact result for the stated lumped model, not an empirical test rule.

Where flight coupling enters

Represent a linear source by its free-interface velocity v_{free} and source impedance Z_s. Connecting a passive load Z_L gives $v = v_{free}Z_s/(Z_s + Z_L)$. A very high-impedance source approaches prescribed motion. A finite support changes motion as the payload reacts against it. Because the impedances are complex, coupling can shift peaks and produce amplification as well as attenuation; there is no universal scalar correction.

A precise limitation follows: if the exact flight motion at every interface degree of freedom were reproduced on an unchanged linear article with all relevant forcing represented, its response would be reproduced. Overtest arises because qualification commonly uses a smoothed or enveloped motion specification and simplified mounting excitation, losing the original source-load relationship.

NASA’s force-limiting literature addresses this impedance mismatch [1,2]. An empirical force curve is one possible approximation to coupled loads, but the test-item resonance alone cannot establish the permissible flight load. The mounting structure and environment must enter that decision.

A 10 kg article can look like 250 kg

Assume m = 10 kg, f_n = 100 Hz and ζ = 0.02. Prescribe a one-sided acceleration PSD S_{aa} = 0.040 g²/Hz over 50-150 Hz. Use g = 9.80665 m/s². Suppose a separate, approved loads assessment has supplied a force amplitude spectral density cap of 100 N/√Hz throughout this illustrative band.

At 100 Hz, $|M_a|$ = 250.20 kg. The input ASD is √0.040 = 0.200 g/√Hz, or 1.9613 m/s²/√Hz. The predicted force ASD is therefore 490.73 N/√Hz. To meet the assumed cap, the input ASD falls to 100/250.20 = 0.3997 m/s²/√Hz = 0.04076 g/√Hz.

$$S_{aa,limited}(f) = \min[S_{aa,target}(f), S_{FF,limit}(f)/|M_a(f)|^2]$$
$$\text{At resonance: } 0.001661 \text{ g}^2/\text{Hz}; \quad 10 \log_{10}(0.001661/0.040) = -13.82 \text{ dB}$$

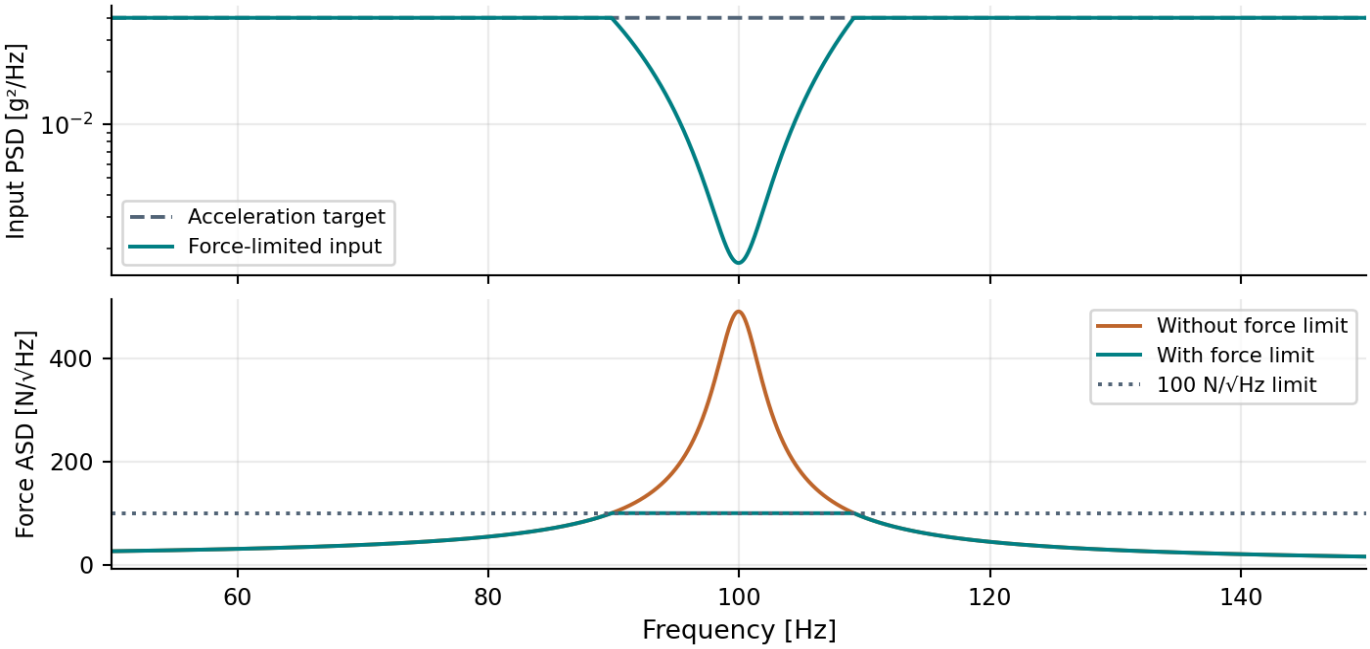


Figure 1. Original SDOF calculation. Both curves use the same article. The upper plot is acceleration PSD; the lower is force ASD. Unit conversion by g² is included when dividing a force PSD by apparent mass squared. This is ideal frequency-by-frequency limiting, not a simulated controller.

The notch depth is a consequence of the assumed force cap and measured or modeled dynamics. It is not permission to reduce every frequency by 13.82 dB. Nor is that narrowband reduction the reduction in overall response RMS: RMS requires integrating the achieved response PSD across the full band.

A real controller has finite resolution, averaging and update time. Close modes, resonance drift or amplitude-dependent damping can change the achieved notch. Preview low-level apparent mass and verify full-level force, response and controller behavior; do not substitute this ideal curve for those observations.

A notch needs an evidence trail

The central review question is: what flight load, with what uncertainty and margin, does this limit represent? A strength allowable alone says what the article can tolerate; it does not show that the test covers flight. Conversely, a low predicted load without model validation can create an attractive but unjustified notch. Keep the environment estimate, qualification margin and hardware protection threshold distinguishable.

The following is an engineering review aid, not a set of requirements copied from a standard. It organizes the decisions needed before accepting a force-limited result.

Decision	Evidence to retain
Define the boundary	Force plane, axis, attachment geometry and hardware configuration match the loads basis.
Establish the cap	Coupled analysis, flight/integrated-test data or a justified empirical method; uncertainty and margin are explicit.
Validate the measurement	Low-level apparent mass, polarity, calibration, preload, bypass paths and mass above the sensors.
Preserve critical responses	Local acceleration/strain, individual mount loads and moments supplement the resultant force where needed.
Accept the achieved test	Actual acceleration and force spectra, response peaks, duration, aborts and configuration changes.

Table 1. Original review aid. Acceptance remains program- and hardware-specific.

Sum forces before taking spectra

For multiple mounts, the resultant is the signed sum of synchronized force histories, or the complex sum at a harmonic frequency. Summing individual force PSDs discards cross terms. It can overestimate a resultant when forces oppose and underestimate it when they reinforce. Two equal, opposite mount forces may produce nearly zero net force but a substantial moment. A net-force limit therefore cannot protect every local load path.

Sensors also measure what lies above them. Adapter inertia can contaminate the inferred article force; a compliant adapter makes a simple real-valued mass subtraction unreliable. NASA-HDBK-7004C discusses these installation and phase-error concerns [1]. Low-frequency apparent mass is a useful sanity check, but correct total mass does not validate the resonant load distribution.

Use the notch to improve test fidelity

Force limiting is most useful when pronounced article resonances interact with a mounting structure whose impedance differs materially from the shaker. It cannot repair an incorrect acceleration environment, an unrepresented acoustic field, or an inadequate test duration. A highly distributed structure may need several force and moment coordinates plus local response limits. Treat each added limit as a coverage decision, not merely a controller setting.

Coherence remains relevant even with accurate force sensors. Kolaini and Kern's 2012 work identifies spatially incoherent mounting excitation as a limitation of traditional force-limit methods [3]. A coherent rigid-table test and a flexible, acoustically excited support can energize modes differently. Matching one summed force spectrum does not establish equivalent modal forces everywhere.

Three takeaways

1. Inspect apparent mass. A compliant 10 kg article can require the force associated with roughly 250 kg at a lightly damped resonance in the stated example.

2. Justify the force cap independently. A notch is credible when its load basis, uncertainty, margin and measurement plane are traceable.

3. Validate the achieved exposure. Retain force, acceleration, local response and duration evidence. Resultant force alone can miss opposing mount loads and moments.

Reflection

A test has a deep notch and every summed-force channel stays below its limit. Does that establish qualification coverage? **Answer:** no. Check the flight basis of the limit, required response and duration coverage, and whether force cancellation or unmeasured moments conceal a critical mode. A well-controlled test is only as defensible as its specified boundary conditions.

Sources, scope and credits

[1] NASA, NASA-HDBK-7004C: Force Limited Vibration Testing, November 30, 2012. Foreword; §§4.2, 5.1-5.6, 6. NASA-authored handbook, public mirror. This edition is engineering guidance, not a universal test requirement or proof of a numerical limit.

[2] T. D. Scharton, Force Limited Vibration Testing Monograph, NASA-RP-1403, May 1997. NASA NTRS. Historical development and theoretical background; its appended draft handbook is not used as a requirement.

[3] A. R. Kolaini and D. L. Kern, A New Approach in Force-Limited Vibration Testing of Flight Hardware, October 15, 2012. NASA NTRS abstract. Spatial coherence limitations.

[4] NASA Webb Telescope Team, NASA Gives the Webb Telescope a Shakedown, February 13, 2017. Photo source and context. Photograph: NASA/Chris Gunn.

Verified October 4, 2026. All numerical results and charts are original illustrative calculations, not measured data or approved test limits. No proprietary program information is used.